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WINDOW KIT OF AIR CONDITIONER

Abstract

A window kit of an air conditioner is provided. The window kit of an air conditioner is configured to be mounted on a window frame connected to a casement window and coupled with an exhaust guide that extends from an exhaust port of a mobile air conditioner. The window kit may include a first installation member configured to be mounted between a rotating end portion of the casement window and the window frame and including an insertion hole into which a nozzle of the exhaust guide is configured to be inserted, and a second installation member coupled to one end portion of the first installation member and mounted on the window frame. The first installation member is configured to have a variable length. The second installation member may include a base part or portion coupled to the first installation member, and a plurality of variable parts or portions detachably connected to an upper end of the base portion.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This application claims priority under 35 U.S.C. § 119 to Korean Application No. KR 10-2024-0032281 filed in Korea on Mar. 6, 2024, whose entire disclosure is hereby incorporated by reference.

BACKGROUND

1. Field

[0002] A window kit for an air conditioner is disclosed herein.

2. Background

[0003] In general, an air conditioner is a cooling/heating system that cools or heats a room by repeatedly suctioning in indoor air, exchanging heat with a low-temperature or high-temperature refrigerant, and then discharging the air into the room. The air conditioner performs a series of cycles including a compressor, an outdoor heat exchanger, an expansion valve, and an evaporator.

[0004] Air conditioners generally include two types, separate air conditioners, in which an indoor unit and an outdoor unit are separated, and integrated air conditioners, in which the indoor unit and the outdoor unit are combined. The separate air conditioner includes an outdoor unit that is mainly installed outdoors (sometimes referred to as the ‘outdoor side’ or ‘heat dissipation side’) and an indoor unit that is mainly installed inside of a building (sometimes referred to as the ‘indoor side’ or ‘heat absorption side’). An outdoor heat exchanger and a compressor are installed in the outdoor unit, and an evaporator is installed in the indoor unit.

[0005] The integrated air conditioner is a type in which an indoor unit and an outdoor unit are combined, and an indoor heat exchanger and an outdoor heat exchanger are provided in one main body, so that the entire air conditioner is installed in one main body. In addition, the integrated air conditioner may be classified into a fixed air conditioner that is fixedly installed on a window, or a window frame, for example, and a mobile air conditioner, a main body of which may be moved to a required space and discharge heat-exchanged air to the outside through an exhaust guide means.

[0006] Among these air conditioners, the mobile air conditioner may be configured to have an exhaust guide means fixed to a window to discharge heat-exchanged air to the outside. The exhaust guide means of the mobile air conditioner may be fixed to the window to ensure stable air discharge.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] Embodiments will be described in detail with reference to the following drawings in which like reference numerals refer to like elements, and wherein:

[0008] FIG. 1 is a perspective view illustrating a state in which an exhaust guide means of a mobile air conditioner is coupled to a window kit of an air conditioner according to an embodiment;

[0009] FIG. 2 is a perspective view illustrating a state in which a window kit of an air conditioner is installed on a casement window according to an embodiment;

[0010] FIG. 3 is an exploded perspective view of a window kit of an air conditioner according to an embodiment;

[0011] FIG. 4 is a plan view of a first installation member of a window kit of an air conditioner according to an embodiment;

[0012] FIG. **5** is a cross-sectional view, taken along line V-V of FIG. **4**;
[0013] FIG. **6** is a perspective view of a second installation member and a third installation member of the window kit of an air conditioner according to an embodiment;
[0014] FIG. **7** is an enlarged perspective view of area A of FIG. **2**;
[0015] FIG. **8** is an enlarged perspective view of area B of FIG. **2**;
[0016] FIG. **9** is a cross-sectional view, taken along line IX-IX of FIG. **1**;
[0017] FIG. **10** is a cross-sectional view, taken along line X-X of FIG. **9**;
[0018] FIG. **11** is an enlarged perspective view of area C in FIG. **2**;
[0019] FIG. **12** is a cross-sectional view, taken along line XII-XII of FIG. **11**;
[0020] FIG. **13** is a cross-sectional view, taken along line XIII-XIII of FIG. **11**;
[0021] FIG. **14** is a view of a modified window kit of an air conditioner according to an embodiment installed on a window of a different size from the window of FIG. **1**;
[0022] FIG. **15** is a view of a window kit of an air conditioner according to another embodiment;
[0023] FIG. **16** is a view of a window kit of an air conditioner according to another embodiment;
and
[0024] FIG. **17** is a cross-sectional view, taken along line XVII-XVII of FIG. **16**.

DETAILED DESCRIPTION

[0025] Hereinafter, embodiments will be described with reference to the drawings. However, the embodiments cannot be said to be limited to the embodiments in which the idea is presented, and other embodiments that are regressive or other embodiments included within the scope may be easily suggested by adding, changing, or deleting other components.

[0026] FIG. **1** is a perspective view illustrating a state in which an exhaust guide means of a mobile air conditioner is coupled to a window kit of an air conditioner according to an embodiment. FIG. **2** is a perspective view illustrating a state in which a window kit of an air conditioner is installed on a casement window according to an embodiment. FIG. **3** is an exploded perspective view of a window kit of an air conditioner according to an embodiment.

[0027] Referring to FIGS. **1** to **3**, air conditioner **1** according to an embodiment may include a main body **10** and an exhaust guide means **20** configured to discharge heat-exchanged air from the main body **10** outside (in particular outdoors). As illustrated, the air conditioner **1** according to an embodiment is an integrated air conditioner in which an indoor unit and an outdoor unit are combined, and in which an indoor heat exchanger and an outdoor heat exchanger may be provided in one main body **10**.

[0028] The air conditioner **1** according to an embodiment may be a mobile air conditioner, that is, the main body **10** may be moved to a required space and discharge heat-exchanged air to the outside through the exhaust guide means **20**. The exhaust guide means **20** of the mobile air conditioner **1** may be fixed to a window for stable air discharge and configured to discharge heat-exchanged air to the outside.

[0029] Components forming a refrigeration cycle may be built into the main body **10**. For example, the main body **10** may be equipped with an evaporator, a heat exchanger that functions as a condenser, a compressor, an expansion device, and a fan/fan motor for air flow. Although not illustrated, the main body **10** may be formed with a suction port through which indoor air is suctioned in and a discharge port through which heat-exchanged air is discharged into a room. Additionally, an exhaust port through which air is discharged to the outdoors may be formed in the main body **10**, and the exhaust guide means **20** may be mounted on the exhaust port.

[0030] The air in a room in which the main body **10** is located may be suctioned in from the suction port, exchange heat with a heat exchanger inside of the main body **10**, and then be discharged into an indoor space through the discharge port. In addition, air that has passed through the heat exchanger and exchanged heat may pass through the exhaust port and be discharged to the outdoor space by the exhaust guide means **20**. This action makes it possible to cool or heat the indoor space.

[0031] The exhaust guide means **20** is configured to guide a flow of exhausted air by connecting

the exhaust port of the main body to a window kit **2** installed on a window frame. The exhaust guide means **20** may include an exhaust pipe **21** forming an air flow path, a nozzle **22** connected to a first installation member **100** at an end portion of the exhaust pipe **21**, a main body connector **23**, and a nozzle connector **24**.

[0032] The exhaust pipe **21** may have a hollow tube shape and be flexible enough to be bent according to a position and height of a window to which the exhaust guide means **20** is connected. Therefore, the exhaust pipe **21** may extend lengthwise and may be made of a flexible plastic, for example. Additionally, the exhaust pipe **21** may be configured to maintain an original shape thereof even when bent as illustrated in the drawing.

[0033] The nozzle **22** may be combined or coupled with the first installation member **100**, which will be described hereinafter, to fix the exhaust guide means **20**, and to allow the discharged air to be completely discharged outdoors. The nozzle **22** may be made of a plastic material, for example, and may be coupled to the first installation member **100**.

[0034] The main body connector **23** may be provided between the main body **10** and the exhaust pipe **21** to guide a lower end of the exhaust pipe **21** to be mounted on the main body **10**. The nozzle connector **24** may be provided between the exhaust pipe **21** and the nozzle **22** to allow the nozzle **22** to be fastened to an upper end of the exhaust pipe **21**.

[0035] The window kit **2** of the air conditioner according to an embodiment may be provided to fix the nozzle **22** of the exhaust guide means **20** to a casement window **3**. The casement window **3** may include a window frame **31** and a window **32**.

[0036] The window kit **2** of the air conditioner according to an embodiment may be installed on the window frame **31** and the window **32**. For example, the window **32** may be configured to open and close an opening of the window frame **31** by opening and closing the window frame **31** through rotation. In one embodiment, the window **32** may include a fixed end portion **32a** forming a rotational axis, a rotating end portion **32b** located on an opposite side of the fixed end portion **32a** and moving in an arc when the window **32** is opened and closed, and a side end portion **32c** that connects the fixing end portion **32a** and the rotating end portion **32b**.

[0037] Correspondingly, the window frame **31** may include a first frame part or portion **31a** corresponding to the fixed end portion **32a**, a second frame part or portion **31b** corresponding to the rotating end portion **32b**, and a third frame part or portion **31c** corresponding to the side end portion **32c**. The window frame **31** may further include a first window frame rib **31d** that protrudes inside of the second frame portion **31b** and a second window frame rib **31e** that protrudes inside of the third frame portion **31c**. The first installation member **100** may be fixed to the first window frame rib **31d**. Second and third installation members **200** and **300** may be fixed to the second window frame rib **31e**.

[0038] The first window frame rib **31d**, illustrated in FIG. **3**, may have a first length **L11**, and the second window frame rib **31e** may have a second length **L21**. Referring to FIG. **14**, which will be described hereinafter, when sizes of the first window frame rib **31d** and the second window frame rib **31e** are changed, it will be explained that window kit **2** according to another embodiment may be structurally changed.

[0039] In this embodiment, the first frame portion **31a** corresponding to the fixed end portion **32a** constitutes an upper end portion of the window frame **31**, and the rotating end portion **32b** is explained based on structure of the window **32** that opens and closes toward an indoor side; however, embodiments are not limited thereto and may also be applied to a window having a different type of casement structure. For example, this embodiment may also be applied to structure in which the first frame portion **31a** constitutes the lower end portion or both side portions of the window frame, and may be also applied to structure of a window in which the rotating end portion **32b** opens and closes toward the outside.

[0040] The window kit **2** of the air conditioner **1** may be fixed to the window frame **31** and the window **32**. The window kit **2** of the air conditioner **1** may include first installation member **100**

fixed to the rotating end portion **32b** of the window **32** and the first window frame rib **31d**, second and third installation members **200**, **300** coupled to both end portions of the first installation member **100** and fixed to the second window frame rib **31e**, and a fixing member **400** configured to fix the first installation member **100** to the rotating end portion **32b** of the window **32** and the first window frame rib **31d**.

[0041] The first installation member **100** may be fixed to the window frame **31** and the window **32** and may be connected to the exhaust guide means **20**. For example, one or a first side of the first installation member **100** may be coupled to the first window frame rib **31d**, and the other or a second side thereof may be coupled to the rotating end portion **32b** of the window **32**.

[0042] The first installation member **100** may include an installation panel **110** to which the exhaust guide means **20** may be connected, and a sliding panel **120** that is coupled to the installation panel **110** and slides in and out in a longitudinal direction of the installation panel **110**. Through the structure of the sliding panel **120** configured to slide in and out with respect to the installation panel **110**, a length of the first installation member **100** may be set in various ways according to a length of the rotating end portion **32b** and the second frame portion **31b** of the installed window **32**.

[0043] An insertion hole **110a** into which the nozzle **22** may be inserted may be formed in the installation panel **110**, and an end portion of the nozzle **22** may penetrate the insertion hole **110a**. Additionally, upper and lower ends of the installation panel **110** may be bent so that the sliding panel **120** may be slidably coupled to the installation panel **110**.

[0044] Structure of the first installation member **100** will be described hereinafter with reference to FIGS. **4** and **5**.

[0045] The second and third installation members **200** and **300** may be coupled to both end portions of the first installation member **100** in an extension direction, respectively. For example, the second installation member **200** may be coupled to one or a first end portion of the first installation member **100**, and the third installation member **300** may be coupled to the other or a second end portion of the first installation member **100**. The second installation member **200** and the third installation member **300** may have a symmetrical structure.

[0046] The second installation member **200** may be fastened to the second window frame rib **31e** on one or a first side through a first fastening member **201**, and the third installation member **300** may be fastened to the second window frame rib **31e** on the other or a second side through a second fastening member **301**.

[0047] Structures of the second and third installation members **200** and **300** will be described hereinafter with reference to FIG. **6**.

[0048] The fixing member **400** may fix the first installation member **100** to the window frame **31** and the window **32**. For example, the first installation member **100** may be fixed to the second frame portion **31b** and the first window frame rib **31d**. The fixing member **400** does not cause damage, such as perforations, to the first window frame rib **31d**, and may fix the first installation member **100** to the first window frame rib **31d**.

[0049] The fixing member **400** included in the window kit **2** of the air conditioner according to an embodiment may be located in a place in which it may be easily fixed in the room. Therefore, the window kit **2** may be easily fixed to or separated from the casement window **3**.

[0050] The fixing member **400** may include a first fixing member **410**, which may be fixed to an inner end of the first installation member **100**, a second fixing member **420** fixed to an outer end of the first installation member **100**, and at the same time, fixed to the rotating end portion of the window **32**, and a restraining member **430** rotatably connected to the first fixing member **410** to restrain or fix the first fixing member **410** to the first window frame rib **31d**. The first fixing member **410** may be coupled to the first installation member **100**, and the restraining member **430** may restrain or fix the first fixing member **410** to the first window frame rib **31d**, and thus, the fixing member **400** according to an embodiment may fix the first installation member **100** to the

first window frame rib **31d** without causing damage, such as perforations, to the first window frame rib **31d**.

[0051] Fixing the first installation member **100** using the fixing member **400** will be described hereinafter with reference to FIGS. **7** and **8**.

[0052] FIG. **4** is a plan view of a first installation member of a window kit of an air conditioner according to an embodiment. FIG. **5** is a cross-sectional view, taken along line V-V of FIG. **4**.

[0053] Referring to FIGS. **4** and **5**, the first installation member **100** may include the installation panel **110** including the insertion hole **110a** into which the nozzle **22** of the exhaust guide means **20** is inserted and the sliding panel **120** that is slid in and out in the longitudinal direction of the installation panel **110**. The installation panel **110** may include a plate-shaped first panel part or portion **111** having a predetermined width and a restraining part or portion **111a** disposed on one side of the first panel portion **111**.

[0054] The first panel portion **111** may form an outer shape of the installation panel **110**, and the insertion hole **110a** may be formed in the first panel portion **111** to allow the nozzle **22** to be inserted therethrough. In the first panel portion **111**, the restraining portion **111a** may be disposed at one end portion connected to the sliding panel **120**. The restraining portion **111a** may include a restraining protrusion **111b** that protrudes downward. The restraining protrusion **111b** of the restraining portion **111a** may be partially inserted into and restrained in a locking groove **123b** of the sliding panel **120**, which will be described hereinafter. Accordingly, an operation of sliding the sliding panel **120** with respect to the installation panel **110** may be performed in a state in which the restraining protrusion **111b** is separated from the locking groove **123b**.

[0055] The sliding panel **120** may be slid in and out along a guide part or portion formed by partially bending upper and lower ends of the installation panel **110**. The sliding panel **120** may include a plate-shaped second panel part or portion **121** having a predetermined width and a locking part or portion **122** formed inside of the second panel portion **121**.

[0056] The second panel portion **121** may form an outer shape of the sliding panel **120**. The sliding panel **120** may be slid in and out with respect to the installation panel **110**, and a width of the second panel portion **121** may be smaller than a width of the first panel portion **111**. The locking portion **122** may be formed inside of the second panel portion **121** to lock the locking protrusion **111b** of the installation panel **110**.

[0057] The locking portion **122** may be formed to correspond to a position of the restraining protrusion **111b** of the installation panel **110**. For example, the locking portion **122** may be formed in or at a central portion of the second panel portion **121**. The locking portion **122** may include a plurality of locking protrusions **123a** arranged in a longitudinal direction of the second panel part **121**. The plurality of locking protrusions **123a** may be spaced apart from each other, and locking groove **123b** may be formed between neighboring locking protrusions **123a**. As described above, the locking protrusion **111b** of the installation panel **110** may be partially inserted into the locking groove **123b** and locked.

[0058] Through the above structure, a user may adjust the length of the first installation member **100** to a desired size. For example, in a state in which the restraining protrusion **111b** of the installation panel **110** is separated from the locking groove **123b** of the sliding panel **120**, the sliding panel **120** may be slid in and out with respect to the installation panel **110**. With this structure, the length of the first installation member **100** may be set to the desired size, and then the restraining protrusion **111b** of the installation panel **110** may be locked and restrained in the locking groove **123b** of the sliding panel **120** and fixed to a set size.

[0059] FIG. **6** is a perspective view of a second installation member and third installation member of the window kit of the air conditioner according to an embodiment. Referring to FIG. **6**, the second installation member **200** and the third installation member **300** of the window kit **2** of the air conditioner according to an embodiment may have structure that is symmetrical to each other. Herein, description is based on the second installation member **200**, but the contents may be

equally applied to the third installation member **300** symmetrically.

[0060] The second installation member **200** may include a base part or portion **210**, a fixing part or portion **220** that fixes the base portion **210** to one end portion of the first installation member **100**, and a variable part or portion **230** detachably connected to the base portion **210**. The base portion **210** may include a fixed base part or portion **210a** fixed to the second window frame rib **31e** and a shielding base part or portion **210b** that shields an open area between the side end portion **32c** and the third frame portion **31c**.

[0061] The fixed base portion **210a** may have a shape corresponding to the second window frame rib **31e**. The fixed base portion **210a** may have a flat bar shape extending in one direction. For example, the fixed base portion **210a** may be disposed on an inner surface of the second window frame rib **31e** and fixed by the fastening member **201**. A fastening hole **210h** through which the fastening member **201** may pass may be formed on one surface of the fixed base portion **210a**. For example, the fastening hole **210h** may be formed at upper and lower end portions of the fixed base portion **210a**; however, embodiments are not limited thereto.

[0062] The shielding base portion **210b** may be formed by bending the fixed base portion **210a** from an inner end to the outside. The shielding base portion **210b** may have a shape corresponding to the open area between the side end portion **32c** and the third frame portion **31c**. The window kit **2** of the air conditioner according to an embodiment may be applied to a casement window, and thus, may have a right triangle flat shape of the shielding base portion **210b**. For example, in the flat shape of the shielding base portion **210b**, a side perpendicular to the base may be a corner portion in contact with the fixed base portion **210a**.

[0063] A fixing part or portion **220** may be provided on an inner surface of the shielding base portion **210b**. The fixing portion **220** may be provided to couple the first installation member **100** and the second installation member **200**. The fixing portion **220** may protrude inward from the inner surface of the shielding base portion **210b**. The fixing portion **220** may include a first fixing rib **221**, and a second fixing rib **222** spaced downward from the first fixing rib **221**. A groove into which the first installation member **100** may be partially inserted may be formed between the first fixing rib **221** and the second fixing rib **222**. For example, one end portion of the installation panel **110** of the first installation member **100** may be partially inserted into and fixed to a groove formed between the first fixing rib **221** and the second fixing rib **222**.

[0064] A coupling structure of the first installation member **100** and the second installation member **200** through the fixing portion **220** will be described hereinafter with reference to FIGS. **11** to **13**.

[0065] A detachable variable part or portion **230** may be disposed on an upper side of the base portion **210**. A plurality of variable parts or portions **230** may be provided and each variable portion **230** may be formed to be partially separable. The second installation member **200** in a state in which the base portion **210** and the variable portion **230** are connected may shield the open area between the side end portion **32c** and the third frame portion **31c**. Accordingly, a size of the second installation member **200** according to an embodiment may be changed to match a vertical size of the window frame **31** and the window **32**. In other words, the second installation member **200** according to an embodiment may be universally applied to window frames **31** and windows **32** of various sizes.

[0066] Hereinafter, as an example, structure of the second installation member **200** including **10** variable portions **230** will be described.

[0067] A first variable part or portion **231** may be disposed at an upper end of the base portion **210** with a first cutout part or portion **231I** interposed therebetween. Along the first cutout portion **231I**, the base portion **210** and the first variable portion **231** may be partially coupled. In other words, the base portion **210** and the first variable portion **231** may be partially separated along the first cutout portion **231I**. Accordingly, the first variable portion **231** and the base portion **210** may be easily separated along the first cutout portion **231I**. For example, when an external force is applied along the first cutout portion **231I** or cut with a cutting tool, the base portion **210** and the first variable

portion **231** may be easily separated. Description of the first cutout portion **231I** may be equally applied to the second to tenth cutout portions **232I** to **240I** described hereinafter.

[0068] The first variable portion **231** may include a first fixed variable part or portion **231a** disposed above the fixed base portion **210a** and a first shielded variable part or portion **231b** disposed above the shielding base portion **210b**. The first fixed variable portion **231a** may have a shape corresponding to the second window frame rib **31e** in a state of being connected to the fixed base portion **210a**. The first fixed variable portion **231a** may have a flat bar shape extending in one direction. The first fixed variable portion **231a** may have a first height **h1**.

[0069] The first fixed variable portion **231a** may be disposed on an inner surface of the second window frame rib **31e** and fixed by the fastening member **201**. A first fastening hole **231h** through which the fastening member **201** may pass may be formed on one surface of the first fixed variable portion **231a**.

[0070] The first shielding variable portion **231b** may be formed by bending the first fixed variable portion **231a** from an inner end to the outside. The first shielding variable portion **231b** may have a shape corresponding to the open area between the side end portion **32c** and the third frame portion **31c** in a state of being connected to the shielding base portion **210b**.

[0071] The second variable portion **232** may be disposed on an upper end of the first variable portion **231** with the second cutout portion **232I** interposed therebetween. The second variable portion **232** and the first variable portion **231** may be easily separated along the second cutout portion **232I**. The second variable portion **232** may include a second fixed variable part or portion **232a** disposed above the first fixed variable portion **231a** and a second shielding variable part or portion **232b** disposed above the first shielding variable portion **231b**.

[0072] The second fixed variable portion **232a** may have a second height **h2**. A second fastening hole **232h** through which the fastening member **201** may pass may be formed on one surface of the second fixed variable portion **232a**.

[0073] A third variable part or portion **233** may be disposed on an upper end of the second variable portion **232** with a third cutout part or portion **233I** interposed therebetween. The third variable portion **233** may include a third fixed variable part or portion **233a** disposed above the second fixed variable portion **232a** and a third shielding variable part or portion **233b** disposed above the second shielding variable portion **232b**. The third fixed variable portion **233a** may have a third height **h3**. A third fastening hole **233h** through which the fastening member **201** may pass may be formed on one surface of the third fixed variable portion **233a**.

[0074] Fourth to tenth variable portions **234** to **240** may be disposed at an upper end of the third variable portion **233** with fourth to tenth cutout portions **234I** to **240I** interposed therebetween. Each of the fourth to tenth variable portions **234** to **240** may include the fourth to tenth fixed variable portions **234a** to **240a** disposed above the third fixed variable portion **233a**, and fourth to tenth shielding variable portions **234b** to **240b** disposed on an upper side of the third shielding variable portion **233b**. Each of the fourth to tenth fixed variable portions **234a** to **240a** may have a fourth to tenth height **h4** to **h10**. For example, the first to tenth heights **h1** to **h10** of the first to tenth fixed variable portions **231a** to **240a** may be set to be the same; however, embodiments are not limited to this and may be set in various ways. Fourth to tenth fastening holes **234h** to **240h** through which the fastening member **201** may pass may be formed on one surface of the fourth to tenth fixed variable portions **234a** to **240a**.

[0075] In addition, description of the second to tenth fixed variable portions **232a** to **240a** and the second to tenth shielding variable portions **232b** to **240b** have been omitted as the description may be applied in the same way as the description of the first fixed variable portion **231a** and the first shielding variable portion **231b**.

[0076] The second installation member **200** may include a front portion **200a** including a fixed base part or portion **210a** and first to tenth fixed variable parts or portions **231a** to **240a** and a side portion **200b** including a shielding base part or portion **210b** and first to tenth shielding variable

parts or portions **231b** to **240b**.

[0077] The window kit **2** for an air conditioner according to an embodiment may further include a first support member **250** that fixes the second installation member **200**. The first support member **250** may have a plate shape that has a predetermined width and extends in a direction in which the second installation member **200** extends.

[0078] The first support member **250** may include a base support part or portion **251** and a plurality of variable support parts or portions **252** connected to an upper end of the base support portion **251**. A support member cutout part or portion **250I** may be formed between the base support portion **251** and the variable support portion **252**. With this structure, the plurality of variable support portions **252** may be easily separated from the base support portion **251**.

[0079] Additionally, the support member cutout portion **250I** may also be formed between each of the plurality of variable support portions **252**. Accordingly, each of the plurality of variable support portions **252** may be easily separated from each other.

[0080] A length of the first support member **250** may be set in various ways by separating the plurality of variable support portions **252**. The length of the first support member **250** may be set to correspond to a length of the second installation member **200**.

[0081] The first support member **250** may be disposed opposite to the fixed base portion **210a** and the first to tenth fixed variable portions **231a** to **240a**. The first support member **250** may include a support member fastening hole **250h** for fixing the second installation member **200** to the second window frame rib **31e**. The support member fastening hole **250h** may be formed in each of the base support portion **251** and the plurality of variable support portions **252**.

[0082] The third installation member **300** may include a base part or portion **310**, a fixing part or portion **320** that fixes the base portion **310** to the end portion of the first installation member **100**, and a variable part or portion **330** detachably connected to the base portion **310**. The base portion **310** may include a fixed base part or portion **310a** fixed to the second window frame rib **31e** and a shielding base part or portion **310b** that shields the open area between the side end portion **32c** and the third frame portion **31c**.

[0083] The fixed base portion **310a** may have a shape corresponding to the second window frame rib **31e**. The fixed base portion **310a** may have a flat bar shape extending in one direction. For example, the fixed base portion **310a** may be disposed on an inner surface of the second window frame rib **31e** and fixed by the fastening member **301**. A fastening hole **310h** through which the fastening member **301** may pass may be formed on one surface of the fixed base portion **310a**. For example, the fastening hole **310h** may be formed at upper and lower ends of the fixed base portion **310a**, respectively; however, embodiments are not limited thereto.

[0084] The shielding base portion **310b** may be formed by bending on an inner end of the fixed base portion **310a** outward. The shielding base portion **310b** may have a shape corresponding to the open area between the side end portion **32c** and the third frame portion **31c**. The window kit **2** of the air conditioner according to an embodiment may be applied to a casement window, and the shielding base portion **310b** may have a right-triangular flat plate shape. For example, in the flat shape of the shielding base portion **310b**, a side perpendicular to the base may be a corner portion in contact with the fixed base portion **310a**.

[0085] A fixing part or portion **320** may be provided on an inner surface of the shielding base portion **310b**. The fixing portion **320** may be provided to couple the first installation member **100** and the third installation member **300**. The fixing portion **320** may protrude inward from the inner surface of the shielding base portion **310b**. The fixing portion **320** may include a first fixing rib **321**, and a second fixing rib **322** spaced downward from the first fixing rib **321**. A groove into which the first installation member **100** is partially inserted may be formed between the first fixing rib **321** and the second fixing rib **322**. For example, one end portion of the sliding panel **120** of the first installation member **100** may be partially inserted into and fixed to a groove formed between the first fixing rib **321** and the second fixing rib **322**.

[0086] A detachable variable part or portion **330** may be disposed on or at an upper side of the base portion **310**. A plurality of variable parts or portions **330** may be provided and the variable portion **330** may be formed to be partially separable. The third installation member **300** in a state in which the base portion **310** and the variable portion **330** are connected may shield the open area between the side end portion **32c** and the third frame portion **31c**. Accordingly, the third installation member **300** according to an embodiment may be changed in size to match a vertical size of the window frame **31** and the window **32**. In other words, the third installation member **300** according to an embodiment may be universally applied to window frames **31** and windows **32** of various sizes. [0087] Hereinafter, structure of the third installation member **300** including 10 variable portions **330** will be described.

[0088] A first variable part or portion **331** may be disposed at an upper end of the base portion **310** with the first cutout portion **331I** interposed therebetween. The first variable portion **331** and the base portion **310** may be easily separated along the first cutout portion **331I**. The first variable portion **331** may include a first fixed variable part or portion **331a** disposed above the fixed base portion **310a** and a first shielded variable part or portion **331b** disposed above the shielding base portion **310b**.

[0089] The first fixed variable portion **331a** may have a shape corresponding to the second window frame rib **31e** in a state of being connected to the fixed base portion **310a**. The first fixed variable portion **331a** may have a flat bar shape extending in one direction. The first fixed variable portion **331a** may have a first height **h1**.

[0090] The first fixed variable portion **331a** may be disposed on an inner surface of the second window frame rib **31e** and fixed by the fastening member **301**. A first fastening hole **331h** through which the fastening member **301** may pass may be formed on one surface of the first fixed variable portion **331a**.

[0091] The first shielding variable portion **331b** may be formed by bending the first fixed variable portion **331a** from the inner end to the outside. The first shielding variable portion **331b** may have a shape corresponding to the open area between the side end portion **32c** and the third frame portion **31c** in a state of being connected to the shielding base portion **310b**.

[0092] A second variable part or portion **332** may be disposed on an upper end of the first variable portion **331** with a second cutout part **332I** interposed therebetween. The second variable portion **332** and the first variable portion **331** may be easily separated along the second cutout portion **332I**. The second variable portion **332** may include a second fixed variable part or portion **332a** disposed above the first fixed variable portion **331a** and a second shielding variable part or portion **332b** disposed above the first shielding variable portion **331b**.

[0093] The second fixed variable portion **332a** may have a second height **h2**. A second fastening hole **332h** through which the fastening member **301** may pass may be formed on one surface of the second fixed variable portion **332a**.

[0094] Third to tenth variable parts or portions **333** to **340** may be disposed on an upper end of the second variable portion **332** with third to tenth cutout parts or portions **333I** to **340I** interposed therebetween. Each of the third to tenth variable portions **333** to **340** may include third to tenth fixed variable parts or portions **333a** to **340a** disposed above the third fixing variable portion **333a** and third to tenth shielding variable parts or portions **333b** to **204b** disposed above the third shielding variable portion **333b**. Each of the third to tenth fixed variable portions **333a** to **340a** may have a third to tenth height **h4** to **h10**. For example, the first to tenth heights **h1** to **h10** of the first to tenth fixed variable portions **331a** to **340a** may be set to be the same; however, embodiments are not limited to this and may be set in various ways.

[0095] Third to tenth fastening holes **333h** to **340h** through which the fastening member **301** may pass may be formed on one surface of the third to tenth fixed variable parts **333a** to **340a**. In addition, description of the second to tenth fixed variable portions **332a** to **340a** and the second to tenth shielding variable portions **332b** to **340b** has been omitted as description may be applied in

the same way as the explanation for the first fixing variable portion **331a** and the first shielding variable portion **331b**, respectively.

[0096] The window kit **2** of an air conditioner according to an embodiment may further include a second support member **350** that fixes the third installation member **300**. The second support member **350** may have substantially a same structure as the second support member **250** described above. For example, the second support member **350** may have a predetermined width and a flat plate shape that extends in a direction in which the third installation member **300** extends.

[0097] The second support member **350** may include a base support part or portion **351** and a plurality of variable support parts or portions **352** connected to an upper end of the base support portion **351**. A support member cutout part or portion **350I** may be formed between the base support portion **351** and the variable support portion **352**. With this structure, the plurality of variable support portions **352** may be easily separated from the base support portion **351**.

[0098] Additionally, the support member cutout portion **350I** may also be formed between each of the plurality of variable support portions **352**. Accordingly, each of the plurality of variable support portions **352** may be easily separated from each other.

[0099] A length of the second support member **350** may be set in various ways by separating the plurality of variable support portions **352**. The length of the second support member **350** may be set to correspond to a length of the third installation member **300**.

[0100] The second support member **350** may be disposed opposite to the fixed base portion **310a** and the first to tenth fixed variable portions **331a** to **340a** of the third installation member **300**. The second support member **350** may include a support member fastening hole **350h** for fixing the third installation member **300** to the second window frame rib **31e**. The support member fastening hole **350h** may be formed in each of the base support portion **351** and the plurality of variable support portions **352**.

[0101] FIG. **7** is an enlarged perspective view of area A of FIG. **2**. FIG. **8** is an enlarged perspective view of area B of FIG. **2**.

[0102] Hereinafter, with reference to FIGS. **7** and **8**, structure for fixing the first installation member **100** to the first window frame rib **31d** and the rotating end portion **32b** of the window **32** using the fixing member **400** will be described. Referring to FIGS. **7** and **8**, the first fixing member **410** may include a first horizontal fixing part or portion **411** coupled to one or a first end of an upper surface of the first installation member **100** and a first vertical fixing part or portion **412** bent upward from one end portion of the first horizontal fixing portions **411**.

[0103] The first horizontal fixing portion **411** may be fastened to the first installation member **100** by a fastening member **411a**. A restraining member **430** may be rotatably coupled to the first vertical fixing portion **412**.

[0104] The second fixing member **420** may include a second horizontal fixing part or portion **421** coupled to the other or a second end of the upper surface of the first installation member **100** and a second vertical fixing part or portion **422** bent upward from one end portion of the second horizontal fixing portion **421**. For example, the second vertical fixing part **422** may include a fastening hole **422h**, and a fastening member **422a** that penetrates the fastening hole **422h** and may be fastened to the rotating end portion **32b** of the window **32**.

[0105] A portion where the fastening member **422a** is fastened to the rotating end portion **32b** may be a portion where a handle of the existing window **32** is fastened. Accordingly, the first installation member **100** may be fixed without forming perforations in the rotating end portion **32b** of the window **32**.

[0106] The restraining member **430** may include a restraining member fixing part or portion **431** coupled to the first vertical fixing portion **412**, a handle portion **432** connected to the restraining member fixing portion **431**, a restraining member **433** that protrudes to one side from the handle portion **432**, and a restraining member rotation part or portion **434** that connects the restraining member fixing portion **431** and the handle portion **432**.

[0107] The restraining member fixing portion **431** may be fastened to and fixed to the first vertical fixing portion **412** by a fastening member **412a**. The handle portion **432** may be rotatably coupled to the restraining member fixing portion **431** by the restraining member rotation portion **434**. In other words, the restraining member rotation portion **434** may constitute a rotational axis of the handle portion **432**.

[0108] The restraining portion **433** may be restrained or fixed to the first window frame rib **31d** as the handle portion **432** rotates. For example, when the restraining portion **433** overlaps the first window frame rib **31d** and the first window frame rib **31d** is located between the restraining portion **433** and the first vertical fixing portion **412**, the first installation member **100** may be fixed to the window frame **31**. When the handle portion **432** rotates so that the restraining portion **433** does not overlap the first window frame rib **31d**, a fixed state of the first installation member **100** to the window frame **31** may be released.

[0109] Through the above structure, the first installation member **100** may be easily fixed to the window frame **31** without causing additional damage, such as perforations, in the window frame **31**. Therefore, when installing the window kit **2** of an air conditioner, it is possible to prevent an aesthetic sense of an outer appearance of the window frame **31** from being deteriorated.

[0110] Hereinafter, with reference to FIGS. **9** and **10**, structure for fixing the second installation member **200** to the second window frame rib **31e** using the first support member **250** will be described.

[0111] FIG. **9** is a cross-sectional view taken along line IX-IX of FIG. **1**. FIG. **10** is a cross-sectional view, taken along line X-X of FIG. **9**.

[0112] Referring to FIGS. **9** and **10**, the second installation member **200** according to an embodiment may be fixed to the second window frame rib **31e** through the first support member **250** without perforations. For example, the front portion **200a** of the second installation member **200** may be fixed on one or a first side of the second window frame rib **31e**, and the first support member **250** may be fixed on the other or a second side thereof. The fastening member **201** may penetrate the front portion **200a** of the second installation member **200** and the support member fastening hole **250h** of the first support member **250**. The fastening member **201** may avoid the second window frame rib **31e** and may not penetrate. The fastening member **201** may be located outside of the end portion of the second window frame rib **31e**. As the fastening holes **250h** of the first support member **250** are formed on both sides, the first support member **250** may be used without distinction between the front surface and the rear surface.

[0113] In a state in which the fastening member **201** is fastened, the second window frame rib **31e** may be located between the front portion **200a** of the second installation member **200** and the first support member **250**. In other words, the second window frame rib **31e** may be partially inserted into a space between the front portion **200a** of the second installation member **200** and the first support member **250**. When the fastening member **201** is sufficiently tightened and fastened, through a binding force between the front portion **200a** of the second installation member **200** and the first support member **250**, the second installation member **200** may be fixed to the second window frame rib **31e**.

[0114] A plurality of fastening members **201** that fastens the front portion **200a** of the second installation member **200** and the first support member **250** may be provided along a longitudinal direction of the first support member **250**. Accordingly, as the fastening member **201** is fastened at a plurality of positions, the second installation member **200** may be more strongly and stably fixed to the second window frame rib **31e**.

[0115] Through the above structure, the second installation member **200** may be easily fixed to the window frame **31** without causing additional damage, such as perforations, in the window frame **31**. Therefore, when installing the window kit **2** of an air conditioner, it is possible to prevent an aesthetic sense of the outer appearance of the window frame **31** from being deteriorated.

[0116] In the above, only the structure for fixing the second installation member **200** to the second

window frame rib **31e** using the first support member **250** has been described, but the above description may be equally applied to structure in which the third installation member **300**, which has a symmetrical structure with the second installation member **200**, is fixed to the second window frame rib **31e** using the second support member **350**.

[0117] Hereinafter, a coupling structure between the first installation member **100** and the second installation member **200** will be described with reference to FIGS. **11** to **13**.

[0118] FIG. **11** is an enlarged perspective view of area C in FIG. **2**. FIG. **12** is a cross-sectional view, taken along line XII-XII of FIG. **11**. FIG. **13** is a cross-sectional view, taken along line XIII-XIII of FIG. **11**.

[0119] Referring to FIGS. **11** to **13**, one end portion of the first panel portion **111** of the first installation member **100** may be partially inserted in a groove formed between the first fixing rib **221** and the second fixing rib **222**. The first fixing rib **221** may include a first fixing rib fastening hole **221h** to which the fastening member **220a** is fastened, and the second fixing rib **222** may include a second fixing rib fastening hole **222h** to which the fastening member **220a** is fastened.

[0120] For example, the installation panel **110** of the first installation member **100** may include a guide part or portion formed by bending downward from both end portions of the first panel portion **111**. The guide portion may provide a path through which the sliding panel **120** is slid in and out. The guide portion may include a first vertical bent part or portion **112** bent downward from both end portions of the first panel portion **111**, a first horizontal bent part or portion **113** bent inward from a lower end portion of the first vertical bent portion **112**, a second vertical bent part or portion **114** bent downward from an inner end portion of the first horizontal bent portion **113**, and a second horizontal bent part or portion **115** bent outward from a lower end portion of the second vertical bent portion **114**.

[0121] The first vertical bent portion **112** may be supported by the first window frame rib **31d**. For example, the first vertical bent portion **112** may be in contact with the first window frame rib **31d** and may be easily aligned in a frontward and rearward direction when fixing the first installation member **100**. In addition, as the first vertical bent portion **112** supports the first window frame rib **31d**, even if the sliding panel **120** slides in and out along the guide portion of the installation panel **110**, it is possible to prevent the sliding panel **120** from interfering with the first window frame rib **31d**.

[0122] FIG. **14** is a view illustrating a modified window kit of an air conditioner according to an embodiment installed on a window of a different size from the window of FIG. **1**. First window frame rib **31d'** and second window frame rib **31e'** included in window frame **31'** illustrated in FIG. **14** may have different lengths from the first window frame rib **31d** and the second window frame rib **31e** illustrated in FIGS. **1** to **3**.

[0123] For example, the first window frame rib **31d'** of the window frame **31'** illustrated in FIG. **14** may have a third length **L11'** smaller than the first length **L11**, and the second window frame rib **31e'** may have a fourth length **L21'** smaller than the second length **L21**.

[0124] The window kit **2'** of the air conditioner in FIG. **14** may include first installation member **100'** and second and third installation members **200'** and **300'** whose lengths are adjustable. The first installation member **100'** may be formed by sliding the sliding panel **120** into and out of the installation panel **110** and then fixing it to correspond to the third length **L11'**. The second installation member **200'** may be formed to correspond to the fourth length **L21'** by partially separating the plurality of variable portions **230**. The structure for fixing the window kit **2'** of the air conditioner to the window frame **31** and the window **32** may be applied in the same manner as described above with reference to FIGS. **1** to **13**.

[0125] In other words, according to the window kit **2** of the air conditioner according to an embodiment, exhaust guide means **20** may be fixed stably to casement window **3** through structure that directly fixes the window kit **2** to the window frame **31**. According to the window kit **2** of the air conditioner according to an embodiment, a size is adjustable and may be installed on casement

windows **3** of various sizes without being limited by the size of the casement window **3**.

[0126] According to the window kit **2** of the air conditioner according to an embodiment, an open area is minimized through the window kit **2** structure that may entirely shield the open area in the casement window **3**, thereby it is possible to prevent air discharged to the outside from being re-introduced into the room through the exhaust guide means **20**.

[0127] According to the window kit **2** of an air conditioner according to an embodiment, the fixing member **400** for fixing the window kit **2** to the window **32** is located in a place in which it may be easily fixed in the room, the window kit **2** may be easily fixed to the casement window **3**.

[0128] According to the window kit **2** of the air conditioner according to an embodiment, the window frame **31** may be fixed without damaging it, thereby preventing the aesthetic sense of the outer appearance of the window frame **31** from being deteriorated.

[0129] Hereinafter, a window kit of an air conditioner according to another embodiment will be described. In the following embodiment, description of the same or similar structure as the already described embodiment has been omitted or simplified, and differences mainly explained.

[0130] FIG. **15** is a view of a window kit for an air conditioner according to another embodiment. Referring to FIG. **15**, window kit **2** of an air conditioner according to this embodiment is different from window kit **2** of an air conditioner according to other embodiments in that adhesive member **260** is applied as a fixing means for fixing when second installation member **200'** is fixed to second window frame rib **31e**.

[0131] The adhesive member **260** may be disposed between a front portion **200a'** of the second installation member **200'** and the second window frame rib **31e**. For example, the adhesive member **260** may be a material having an adhesive strength or a separate member having an adhesive strength on both sides, such as double-sided tape. In this embodiment, first support member **250'** and the second window frame rib **31e** may also be provided with an adhesive member **260**; however, embodiments are not limited to this, and the first support member **250'** may be omitted in some cases.

[0132] FIG. **16** is a view of a window kit for an air conditioner according to another embodiment. FIG. **17** is a cross-sectional view, taken along line XVII-XVII of FIG. **16**.

[0133] Referring to FIGS. **16** and **17**, window kit **2** of an air conditioner according to this embodiment is different from the window kit **2** of an air conditioner according to other embodiments in that restraining member **430** is omitted and first installation member **100** is fixed to the first window frame rib **31d** by first fixing member **140'**. The first fixing member **410'** may include a first horizontal fixing part or portion **411** coupled to one end of the upper surface of the first installation member **100**, and a first vertical fixing part or portion **412'** bent downward from one end portion of the first horizontal fixing portion **411**. The first vertical fixing portion **412'** may be fixed to the first window frame rib **31d** by a fastening member **412a'**. The first vertical bent portion **112** of the first installation member **100** may be supported on the first window frame rib **31d**.

[0134] An end portion of the fastening member **412a'** fastened through the first vertical fixing portion **412'** and the first window frame rib **31d** may support the first horizontal bent portion **113** upward. With this structure, the first installation member **100** may be more stably fixed to the first window frame rib **31d**.

[0135] Embodiments disclosed herein provide a window kit for fixing an exhaust guide means of a mobile air conditioner to a casement window, in particular, a window kit that may be installed on a casement window.

[0136] Embodiments disclosed herein also provide a window kit for fixing an exhaust guide means of a mobile air conditioner that may be universally installed in casement windows of various sizes.

[0137] Embodiments disclosed herein further provide a window kit that may prevent air discharged to the outside through the exhaust guide means from re-entering a room by minimizing an open area in the casement window to which the exhaust guide means of the mobile air conditioner is

fixed.

[0138] Embodiments disclosed herein furthermore provide a window kit for fixing an exhaust guide means of a mobile air conditioner that may be easily fixed to a casement window.

[0139] Embodiments disclosed herein additionally provide a window kit for fixing an exhaust guide means of a mobile air conditioner that may be fixed to a window frame without damaging the window frame.

[0140] A window kit of an air conditioner according to an embodiment, which is mounted on a window frame connected to a casement window and is coupled with an exhaust guide extending from an exhaust port of a mobile air conditioner, may include a first installation member mounted between a rotating end portion of the casement window and the window frame and including an insertion hole into which a nozzle of the exhaust guide is inserted, and a second installation member coupled to one end portion of the first installation member and mounted on the window frame, in which the first installation member is configured to have a variable length. The second installation member may include a base part or portion coupled to the first installation member, and a plurality of variable parts or portions detachably connected to an upper end of the base part.

[0141] The first installation member may include an installation panel mounted on the rotating end portion of the window and the window frame, and a sliding panel connected to the installation panel so as to slide in and out. The sliding panel may include a plurality of locking protrusions arranged in one direction, and a locking groove may be formed between neighboring locking protrusions of the plurality of locking protrusions. The installation panel may include a restraining protrusion inserted into the locking groove and configured to set a length of the first installation member.

[0142] The window frame may include a first window frame rib on which the first installation member may be mounted and which extends in one direction, and a second window frame rib which extends in a direction intersecting the first window frame rib and on which the second installation member may be mounted. The window kit of an air conditioner may further include a support member disposed to face one or a first surface of the second window frame rib. The second installation member may include a front portion facing the other or a second surface of the second window frame rib, and a side portion bent from one end of the front portion. The front portion and the support member may be coupled to each other by a fastening member.

[0143] The second window frame rib may be disposed between the front portion and the support member. The fastening member may be disposed on or at an outside of the second window frame rib.

[0144] The base part may include a fixed base part or portion that faces one or a first surface of the second window frame rib, and a shielding base part or portion bent from one end of the fixed base part. The variable part may include a fixed variable part or portion connected to an upper end of the fixed base part and facing the one surface of the second window frame rib, and a shielding variable part or portion connected to an upper end of the shielding base part and bent from one end of the fixed variable part.

[0145] The second installation member may include a first cutout part or portion disposed between the fixed base part and the fixed variable part, partially coupled, and thus formed to facilitate separation. The second installation member may include a second cutout part or portion disposed between the shielding base part and the shielding variable part, partially coupled, and thus formed to facilitate separation. The second cutout part may be connected to the first cutout part.

[0146] The window kit of an air conditioner may further include a first fixing member coupled to an upper surface of the first installation member, and a restraining member rotatably connected to the first fixing member and configured to be restrained by the first window frame rib as rotating. The first fixing member may include a horizontal fixing part or portion coupled to the upper surface of the first installation member, and a vertical fixing part or portion bent upward from one end portion of the horizontal fixing part. The restraining member may be coupled to the vertical

fixing part.

[0147] The restraining member may include a handle portion connected to the vertical fixing part and configured to rotate, and a restraining portion configured to overlap or not overlap the first window frame rib according to rotation of the handle portion. When the restraining portion and the first window frame rib overlap, the restraining member may restrain so that the first installation member is fixed to the first window frame rib.

[0148] The second installation member may include a front portion that faces one surface of the second window frame rib, and a side portion bent from one end of the front portion. An adhesive member may be disposed between the front portion and the second window frame rib.

[0149] The casement window may further include a fixed end portion that forms a rotational axis of the rotation end portion, and a side end portion that connects the rotating end portion and the fixed end portion. The second installation member may shield an area between the side end portion and the second window frame rib, and the first installation member may shield an area between the rotating end portion and the first window frame rib.

[0150] An inner end portion of the first installation member may be supported by the first window frame rib.

[0151] The second installation member may further include a fixing part or portion that protrudes from the inner surface of the base part. The fixing part may include a first fixing rib, and a second fixing rib spaced below the first fixing rib. The first installation member may be partially inserted into a groove formed between the first fixing rib and the second fixing rib.

[0152] The second installation member may include a first cutout part or portion disposed between the base part and the plurality of variable parts, partially coupled, and thus formed to facilitate separation. The second installation member may further include a second cutout part or portion disposed between adjacent variable parts of the plurality of variable parts, partially coupled, and thus formed to facilitate separation.

[0153] At least the following advantages may be expected from the window kit of an air conditioner according to the embodiment.

[0154] According to the window kit of an air conditioner according to an embodiment, the exhaust guide means may be stably fixed to the casement window through structure in which the window kit is directly fixed to the window frame.

[0155] According to the window kit for an air conditioner according to an embodiment, a size is adjustable and may be installed on casement windows of various sizes without being limited by the size of the casement window.

[0156] According to the window kit of an air conditioner according to an embodiment, an open area is minimized through a window kit structure that may entirely shield the open area in the casement window, and thus the air discharged to the outside through the exhaust guide means may be prevented from re-entering the room.

[0157] According to the window kit of the air conditioner according to the embodiment, the fixing means for fixing the window kit to the window is located in a place in which it can be easily fixed in the room, so the window kit may be easily fixed to the casement window.

[0158] According to the window kit for an air conditioner according to an embodiment, it can be fixed to the window frame without damaging the window frame, thereby preventing aesthetics of an outer appearance of the window frame from being deteriorated.

[0159] It will be understood that when an element or layer is referred to as being “on” another element or layer, the element or layer can be directly on another element or layer or intervening elements or layers. In contrast, when an element is referred to as being “directly on” another element or layer, there are no intervening elements or layers present. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

[0160] It will be understood that, although the terms first, second, third, etc., may be used herein to describe various elements, components, regions, layers and/or sections, these elements,

components, regions, layers and/or sections should not be limited by these terms. These terms are only used to distinguish one element, component, region, layer or section from another region, layer or section. Thus, a first element, component, region, layer or section could be termed a second element, component, region, layer or section without departing from the teachings of the present invention.

[0161] Spatially relative terms, such as “lower”, “upper” and the like, may be used herein for ease of description to describe the relationship of one element or feature to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation, in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “lower” relative to other elements or features would then be oriented “upper” relative to the other elements or features. Thus, the exemplary term “lower” can encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

[0162] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0163] Embodiments are described herein with reference to cross-section illustrations that are schematic illustrations of idealized embodiments (and intermediate structures). As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, embodiments should not be construed as limited to the particular shapes of regions illustrated herein but are to include deviations in shapes that result, for example, from manufacturing.

[0164] Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

[0165] Any reference in this specification to “one embodiment,” “an embodiment,” “example embodiment,” etc., means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the invention. The appearances of such phrases in various places in the specification are not necessarily all referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with any embodiment, it is submitted that it is within the purview of one skilled in the art to effect such feature, structure, or characteristic in connection with other ones of the embodiments.

[0166] Although embodiments have been described with reference to a number of illustrative embodiments thereof, it should be understood that numerous other modifications and embodiments can be devised by those skilled in the art that will fall within the spirit and scope of the principles of this disclosure. More particularly, various variations and modifications are possible in the component parts and/or arrangements of the subject combination arrangement within the scope of the disclosure, the drawings and the appended claims. In addition to variations and modifications in the component parts and/or arrangements, alternative uses will also be apparent to those skilled in the art.

Claims

1. A window kit of an air conditioner, which is configured to be mounted on a window frame connected to a casement window and coupled with an exhaust guide that extends from an exhaust port of a mobile air conditioner, the window kit comprising: a first installation member configured to be mounted between a rotating end portion of the casement window and the window frame and including an insertion hole into which a nozzle of the exhaust guide is inserted; and a second installation member coupled to one end portion of the first installation member and mounted on the window frame, wherein the first installation member is configured to have a variable length, and wherein the second installation member includes: a base portion coupled to the first installation member; and a plurality of variable portions detachably connected to an upper end of the base portion.
2. The window kit of claim 1, wherein the first installation member includes: an installation panel configured to be mounted on the rotating end portion of the casement window and on the window frame; and a sliding panel connected to the installation panel so as to slide in and out.
3. The window kit of claim 2, wherein the sliding panel includes a plurality of locking protrusions arranged in one direction, wherein a locking groove is formed between neighboring locking protrusions of the plurality of locking protrusions, and wherein the installation panel includes a restraining protrusion inserted into the locking groove and configured to set a length of the first installation member.
4. The window kit of claim 1, wherein the window frame includes: a first window frame rib on which the first installation member is configured to be mounted and which extends in one direction; and a second window frame rib that extends in a direction intersecting the first window frame rib and on which the second installation member is configured to be mounted.
5. The window kit of claim 4, further comprising: a support member disposed to face a first surface of the second window frame rib, wherein the second installation member includes: a front portion that faces a second surface of the second window frame rib; and a side portion bent from one end of the front portion, and wherein the front portion and the support member are coupled to each other by a fastening member.
6. The window kit of claim 5, wherein the second window frame rib is disposed between the front portion of the second installation member and the support member, and wherein the fastening member is disposed at an outside of the second window frame rib.
7. The window kit of claim 4, wherein the base portion includes: a fixed base portion that faces the first surface of the second window frame rib; and a shielding base portion bent from one end of the fixed base portion, and wherein the variable portion includes: a fixed variable portion that is connected to an upper end of the fixed base portion and faces the first surface of the second window frame rib; and a shielding variable portion connected to an upper end of the shielding base portion and bent from one end of the fixed variable portion.
8. The window kit of claim 7, wherein the second installation member further includes: a first cutout portion disposed between the fixed base portion and the fixed variable portion, partially coupled, and thus formed to facilitate separation.
9. The window kit of claim 8, wherein the second installation member further includes: a second cutout portion disposed between the shielding base portion and the shielding variable portion, partially coupled, and thus formed to facilitate separation, and wherein the second cutout portion is connected to the first cutout portion.
10. The window kit of claim 4, further comprising: a first fixing member coupled to an upper surface of the first installation member; and a restraining member rotatably connected to the first fixing member and configured to be restrained by the first window frame rib from rotating, wherein the first fixing member includes: a horizontal fixing portion coupled to an upper surface of the first

installation member; and a vertical fixing portion bent upward from one end portion of the horizontal fixing portion, and wherein the restraining member is coupled to the vertical fixing portion.

11. The window kit of claim 10, wherein the restraining member includes: a handle portion connected to the vertical fixing portion and configured to rotate; and a restraining portion configured to overlap or not overlap the first window frame rib according to rotation of the handle portion.

12. The window kit of claim 11, wherein, when the restraining portion and the first window frame rib overlap, the restraining member is restrained so that the first installation member is fixed to the first window frame rib.

13. The window kit of claim 4, wherein the second installation member further includes: a front portion that faces one surface of the second window frame rib; and a side portion bent from one end of the front portion, and wherein an adhesive member is disposed between the front portion and the second window frame rib.

14. The window kit of claim 4, wherein the casement window further includes: a fixed end portion that forms a rotational axis of the rotation end portion; and a side end portion that connects the rotating end portion and the fixed end portion, wherein the second installation member shields an area between the side end portion and the second window frame rib, and wherein the first installation member shields an area between the rotating end portion and the first window frame rib.

15. The window kit of claim 4, wherein an inner end portion of the first installation member is supported by the first window frame rib.

16. The window kit of claim 1, wherein the second installation member further includes a fixing portion that protrudes from an inner surface of the base portion, wherein the fixing portion includes: a first fixing rib; and a second fixing rib spaced below the first fixing rib, and wherein the first installation member is partially inserted into a groove formed between the first fixing rib and the second fixing rib.

17. The window kit of claim 1, wherein the second installation member includes: a first cutout portion disposed between the base portion and the plurality of variable portions, partially coupled, and thus formed to facilitate separation.

18. The window kit of claim 17, wherein the second installation member further includes a second cutout portion disposed between adjacent variable portions of the plurality of variable portions, partially coupled, and thus formed to facilitate separation.

19. An air conditioner comprising the window kit of claim 1.

20. A window kit of an air conditioner, which is configured to be mounted on a window frame connected to a casement window and coupled with an exhaust guide that extends from an exhaust port of a mobile air conditioner, the window kit comprising: a first installation member configured to be mounted between the casement window and the window frame and including an insertion hole into which a nozzle of the exhaust guide is inserted, the first installation member including an installation panel configured to be mounted on the casement window and the window frame, and a sliding panel configured to slide with respect to the installation panel to adjust a length of the first installation panel; and a second installation member coupled to one end portion of the first installation member and mounted on the window frame, the second installation member including a base portion coupled to the first installation member, and a plurality of variable portions detachably connected to an upper end of the base portion.

21. The window kit of claim 20, wherein the second installation member includes: a first cutout portion disposed between the base portion and the plurality of variable portions, partially coupled, and thus formed to facilitate separation, wherein the second installation member further includes a second cutout portion disposed between adjacent variable portions of the plurality of variable

portions, partially coupled, and thus formed to facilitate separation.

22. An air conditioner comprising the window kit of claim 20.
